# STAT 210A Final Cheat sheet

**Loss Function:** $L(\theta, d)$, measures disutility of guessing $g(\theta)=d$

↑ Convexity Loss means in terms of $d$

**Risk Function:** $R(\theta; \delta(\cdot)) = \mathbb{E}_\theta[L(\theta, \delta(X))]$

↳ Ex: risk for square error loss is $MSE(\theta; \delta(\cdot)) = \mathbb{E}_\theta[(\delta(X) - g(\theta))^2]$

**Inadmissibility:** $\delta$ inadmissible if $\exists\, \delta^*$ st. $\begin{cases} R(\theta; \delta^*) \le R(\theta, \delta) & \text{for all } \theta \\ R(\theta; \delta^*) < R(\theta, \delta) & \text{for some } \theta \end{cases}$

↳ $\delta^*$ **strictly dominates** $\delta$

**Average-Case Risk:** $\int_\Theta R(\theta; \delta)\, d\pi(\theta)$ for some prior measure $\pi$

If $\pi$ is prob. meas. $\;= \mathbb{E}_{\theta\sim\pi}[R(\theta; \delta)]$ (minimizing results in Bayes Estimator)

**Worst-Case Risk:** $\sup_\theta R(\theta; \delta)$ (minimizing results in Minimax Estimator)

## Exponential Families

**s-param Expo Family:** $\mathcal{P} = \{P_\eta : \eta \in \Xi\}$ w/ densities $p_\eta$ of form:

$$p_\eta(x) = e^{\eta' T(x) - A(\eta)} h(x),$$

where

- $T: \mathcal{X} \to \mathbb{R}^s$ **sufficient statistic**
- $h: \mathcal{X} \to \mathbb{R}$ **carrier/base density**
- $\eta \in \Xi \subseteq \mathbb{R}^s$ **natural parameter**
- $A: \mathbb{R}^s \to \mathbb{R}$ **log-partition function**

convex!

$^*A(\eta) = \log\left[\int_{\mathcal{X}} e^{\eta' T(x)} h(x)\, d\mu(x)\right] \le \infty$

$^*$**Convexity:** $f(x)$ convex if $\forall x_1, x_2 \in \mathcal{X}, \forall \alpha \in [0,1]$ $f(\alpha x_1 + (1-\alpha)x_2) \le \alpha f(x_1) + (1-\alpha) f(x_2)$

↳ Can interpret as tilting $h(x)$ towards $T(x)$ (via $e^{\eta' T(x)}$) and then renormalizing via $e^{-A(\eta)}$.

Convex b/c $A(\eta)$ convex!

**Natural Parameter Space:** $\Xi_1 = \{\eta : A(\eta) < \infty\}$, normalizable $p_\eta$

**Differential Identities:** $A(\eta)$ has all partial derivatives on $\Xi_1$ by Keener Thm 2.4

(1st) $\frac{\partial A}{\partial \eta_j}(\eta) = \mathbb{E}_\eta[T_j(X)]$, $\nabla A(\eta) = \mathbb{E}_\eta[T(X)] \in \mathbb{R}^s$

(2nd) $\frac{\partial A}{\partial \eta_j \partial \eta_k}(\eta) = Cov_\eta(T_j, T_k)$, $\nabla^2 A(\eta) = Var_\eta(T(X)) \in \mathbb{R}^{s\times s}$ ← $J_1(\eta) = \frac{1}{n} J(\theta)$

**Moment Generating Function:** $M_\eta^T(u) = e^{A(\eta+u) - A(\eta)}$ (MGF of $T(X)$ when $X \sim P_\eta$)

↳ kth order moment of $T(X)$ $= \frac{\partial^k}{\partial \eta^k}(e^{A(\eta)}) \cdot \frac{1}{e^{A(\eta)}}$

**Cumulant Generating Function:** $K_\eta^T(u) = \log M_\eta^T(u) = A(\eta+u) - A(\eta)$ (CGF of $T(X)$)

## Sufficiency

Model w/ $p_\theta(x)$ wrt meas. $\mu$

↳ **Definition:** $T(X)$ sufficient for $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ if $P_\theta(X \mid T)$ independent of $\theta$

**Sufficiency Principle:** if $T(X)$ sufficient for $\mathcal{P}$, then any stat procedure should depend on $X$ only thru $T(X)$. $\theta \to T(X) \to X$, $\to \tilde{X}$ (just as good as X!)

**Factorization Theorem:** $T(X)$ sufficient for model $\mathcal{P}$ iff $\exists\, g_\theta(t), h(x)$ s.t.

$$p_\theta(x) = g_\theta(T(x))\, h(x)$$

for $\mu$-almost-every $x$: $\mu(\{x : p_\theta(x) \ne g_\theta(T(x)) h(x)\}) = 0$

**Order Statistics:** $(X_{(i)})_{i=1}^n$ ($X_{(k)}$ = kth smallest) is sufficient

**Empirical Distribution:** $\hat{P}_n(\cdot) = \frac{1}{n}\sum_{i=1}^n \delta_{X_i}(\cdot)$ is sufficient, where $\delta_{X_i}(A) = 1\{X_i \in A\}$

**Minimal Sufficiency:** $T(X)$ minimal sufficient if (1) $T(X)$ sufficient; and (2) $\forall$ other sufficient $S(X)$, $T(X) = f(S(X))$ for some $f$.

↳ $^*$If $T(X)$ sufficient and $T(X) = f(S(X))$, then $S(X)$ sufficient.

**Likelihood Function:** $Lik(\theta; X) = p_\theta(x)$

↳ **Log-likelihood:** $\ell(\theta; X) = \log Lik(\theta; X)$

↳ Likelihood up to scaling (or $\ell$ up to vert shift) is minimal sufficient!

↳ Likelihood ratios $\left(\frac{Lik(\theta_1; X)}{Lik(\theta_2; X)}\right)_{\theta_1, \theta_2 \in \Theta}$

**Minimal Sufficiency via Likelihood:** $T(X)$ minimal sufficient if

(1) $T(X)$ sufficient

(2) $T(X)$ can be recovered from likelihood shape → $Lik(\cdot; x) \propto Lik(\cdot; y) \Rightarrow T(x) = T(y)$ or equivalently, $\ell(\cdot; x) - \ell(\cdot; y) = const(x,y) \Rightarrow T(x) = T(y)$

**Minimal Sufficiency of Expo Fam:** $T(X)$ minimal if $Span\{\eta(\theta_1) - \eta(\theta_2) : \theta_1, \theta_2 \in \Theta\} = \mathbb{R}^s$

## Completeness

Strat for completeness proofs: Show two things are a.s. equal by showing they have equal expectation

↳ **Definition:** $T(X)$ is complete for $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ if, for any measurable $f$,

$$\mathbb{E}_\theta[f(T(X))] = 0 \;\forall\theta \Rightarrow f(T) \overset{a.s.}{=} 0 \;\;\forall\theta$$

**Full-rank Expo Fam:** $\mathcal{P}$ is full-rank if $T(X)$ satisfies no linear constraint ($\nexists\, \beta \ne 0, \alpha : \beta' T(X) \overset{a.s.}{=} \alpha$), and $\Xi$ contains an open set.

↳ If $\mathcal{P}$ is not full-rank, we say it is **curved**

↳ If $T(X)$ satisfies linear constraint, $\mathcal{P}$ might still be full-rank for lower-dim suff. stat.

**Theorem:** If $\mathcal{P}$ full-rank, then $T(X)$ is complete sufficient

**Theorem:** If $T(X)$ complete sufficient for $\mathcal{P}$, then $T(X)$ minimal

## Ancillarity

↳ **Definition:** $V(X)$ ancillary for $\mathcal{P}$ if its distro independent of $\theta$

**Basu's Theorem:** If $T(X)$ complete sufficient and $V(X)$ ancillary for $\mathcal{P}$, then $V(X) \perp\!\!\!\perp T(X)$ for all $\theta \in \Theta$

↳ If can't verify Thm's hypotheses for one family, try another!

## Unbiased Estimation

↳ **Require:** $\mathbb{E}_\theta[\delta(X)] = g(\theta)$ $\forall\theta \in \Theta$. If $T(X)$ complete sufficient, then there exists at most one unbiased $\delta^*(T(X))$, which uniformly minimizes risk for any convex loss function

**Jensen's:** If $f$ convex then $f(\mathbb{E}[X]) \le \mathbb{E}[f(X)]$ $\forall$ r.v. $X$

**Rao-Blackwell Theorem:** $T(X)$ sufficient, $\delta(X)$ estimator, $\bar{\delta}(T(X)) = \mathbb{E}[\delta(X) \mid T(X)]$ ↖ No $\theta$!

↳ Recipe to improve any $\delta(X)$ that violates suff. principle

Then, we have: $L(\theta, d)$ convex $\Rightarrow R(\theta; \bar{\delta}) \le R(\theta; \delta)$

$L(\theta, d)$ strictly convex $\Rightarrow R(\theta; \bar{\delta}) < R(\theta; \delta)$ (unless $\delta(X) \overset{a.s.}{=} \bar{\delta}(T(X))$ $\forall\theta$)

$^*\bar{\delta}(T)$ = Rao-Blackwellization of $\delta(X)$

**U-estimable:** $g(\theta)$ U-estimable if $\exists\, \delta(X)$ s.t. $\mathbb{E}_\theta[\delta(X)] = g(\theta)$ $\forall\theta$

**Uniform Min Variance Unbiased (UMVU):** $\delta(X)$ UMVU if $\forall$ unbiased $\tilde{\delta}$, $Var_\theta(\delta(X)) \le Var_\theta(\tilde{\delta}(X))$ $\forall\theta \in \Theta$

Use power series on $\mathbb{E}_\theta \delta(T)$ and match terms

**Finding UMVUE:** 1) Find any unbiased estimator based on complete suff. $T(X)$

2) Find any unbiased estimator at all, then R-B'ize it (i.e. $\bar{\delta}(T)$)

## Score

→ **Definition:** $\nabla \ell(\theta; X)$, thought of as "local complete suff. stat."

**Differential Identities:** $\mathbb{E}_\theta[\nabla \ell(\theta; X)] = 0$, $\underbrace{Var_\theta[\nabla \ell(\theta; X)]}_{J(\theta)} = \mathbb{E}_\theta[-\nabla^2 \ell(\theta; X)]$ ← Note all $\theta$ same

$J(\theta)$: **Fisher Information** (squared rate of change of distro wrt param)

No unbiased estimator can have smaller variance than $\nabla g(\theta)' J(\theta)^{-1} \nabla g(\theta)$

**Cramer-Rao Lower-Bound:** Let $g(\theta) = \mathbb{E}_\theta[\delta(X)]$, then:

1-param: $Var_\theta(\delta) \cdot Var_\theta(\dot{\ell}(\theta, X)) \ge Cov_\theta(\delta, \dot{\ell}(\theta; X))^2 \Rightarrow Var_\theta(\delta) \ge \frac{\dot{g}(\theta)^2}{J(\theta)}$

Multivar: $\theta \in \mathbb{R}^d$, $g(\theta), \delta(X) \in \mathbb{R}$: $Var_\theta(\delta) \ge \nabla g(\theta)' J(\theta)^{-1} \nabla g(\theta)$

**Efficiency:** (of unbiased est) $eff_\theta(\delta) = \underbrace{CRLB / Var_\theta(\delta)}_{Corr_\theta^2(\delta, \nabla \ell(\theta; X))} \le 1$, say $\delta(X)$ **efficient** if $eff_\theta(\delta) = 1$ $\forall\theta$ $\Leftrightarrow$ $Corr_\theta^2(\delta, \dot{\ell}(\theta)) = 1$ $\forall\theta$

## Bayes Estimation

**Bayes Risk:** $r_\Lambda(\delta) = \int_\Theta R(\theta; \delta)\, d\Lambda(\theta) = \mathbb{E}_{\theta\sim\Lambda}[\overbrace{R(\theta; \delta)}^{\mathbb{E}[L(\theta; \delta(X)) \mid \theta]}] = \mathbb{E}_{(\theta, X)}[L(\theta, \delta(X))]$ ← $\theta \sim \Lambda$, $X \mid \theta \sim P_\theta$

**Densities:** prior $\lambda(\theta)$, likelihood $p_\theta(x)$, joint density $\lambda(\theta) p_\theta(x)$, marginal density $q(x) = \int_\Theta \lambda(\theta) p_\theta(x)\, d\theta$, posterior density $\lambda(\theta \mid x) = \frac{\lambda(\theta) \cdot p_\theta(x)}{q(x)}$

**Bayes Estimator:** $\delta_\Lambda(x) \in \arg\min_d \mathbb{E}[L(\theta, d) \mid X = x]$ a.e. (w.p. 1)

↳ Ex: if $L(\theta, d) = (g(\theta) - d)^2$ then $\delta_\Lambda(x) = \mathbb{E}[g(\theta) \mid X = x]$

↳ Ex: if $L(\theta, d) = w(\theta)(g(\theta) - d)^2$ then $\delta_\Lambda(x) = \frac{\mathbb{E}[w(\theta) g(\theta) \mid X]}{\mathbb{E}[w(\theta) \mid X]}$

**Conjugate Prior:** prior is conjugate to likelihood if posterior from same family

↳ Ex:

| Likelihood | Prior |
|---|---|
| $X_i \mid \theta \sim Binomial(n, \theta)$ | $\theta \sim Beta(\alpha, \beta)$ |
| $X_i \mid \theta \sim N(\theta, \sigma^2)$ | $\theta \sim N(\mu, \tau^2)$ |
| $X_i \mid \theta \sim Pois(\theta)$ | $\theta \sim Gamma(\nu, s)$ |

**Empirical Bayes:** Common in HBM

$\zeta \sim \lambda(\zeta)$, $\theta_i \mid \zeta \overset{iid}{\sim} \pi_\zeta(\theta)$, $X_i \mid \zeta, \theta \overset{\perp\!\!\!\perp}{\sim} p_{\theta_i}(x)$

↳ Hybrid Approach: treat $\zeta$ as fixed

1. Estimate $\zeta$ based on observed data
2. Plug in $\zeta$ as though it were known

## Hierarchical Bayes

*Full power of Bayes realized in big, complex problems w/ repeat structure so we can pool info

Ex. Predict a batter's "true" batting average from $n_i$ at-bats. $X_i$ = # of hits $\sim \text{Bin}(n_i, \theta_i)$

Pool info across players via hierarchical model (HBM): $\alpha, \beta \sim \lambda_0(\alpha,\beta)$, $\theta_i \mid \alpha, \beta \overset{iid}{\sim} \text{Beta}(\alpha,\beta)$, $X_i \mid \theta_i \overset{\perp}{\sim} \text{Bin}(n_i, \theta_i)$

$\mathbb{E}[\theta_i \mid X] = \mathbb{E}[\mathbb{E}[\theta_i \mid X, \alpha, \beta] \mid X] = \mathbb{E}\left[\frac{X_i+\alpha}{n_i+\alpha+\beta} \mid X\right]$ — = fixed, = sampled $\sim \lambda(\alpha,\beta \mid X)$

**Gaussian Hierarchical Model:** $\tau^2 \sim \lambda_0$, $\theta_i \mid \tau^2 \overset{iid}{\sim} N(0,\tau^2)$, $X_i \mid \tau^2, \theta \overset{\perp}{\sim} N(\theta_i, 1)$

Posterior Mean $\rightarrow$ $\delta(X_i) = \mathbb{E}[\mathbb{E}[\theta_i \mid X, \tau^2] \mid X] = \mathbb{E}[\tau^2/(1+\tau^2) \cdot X_i \mid X] = \mathbb{E}[\tau^2/(1+\tau^2) \mid X] \cdot X_i$

Linear Shrinkage Estimator $\rightarrow$ $\delta(X_i) = (1 - \mathbb{E}[\zeta \mid X])X_i$ where $\zeta(\tau^2) = 1/(1+\tau^2)$ $\leftarrow$ shrinkage factor

## Markov Chain Monte Carlo

$*\ \lambda(\theta \mid x) = \frac{p_\theta(x)\lambda(\theta)}{\int_\Omega p_\zeta(x)\lambda(\zeta)d\zeta}$ $\leftarrow$ nice to compute, $\leftarrow$ often intractable

↳ Compute Strat: use MC w/ stationary dist $\propto p_\theta(x)\lambda(\theta)$, run it to approximate samples from $\lambda(\theta \mid x)$

↳ Definition: sequence of rvs $X^{(0)}, X^{(1)}, \dots$ s.t. $X^{(0)} \sim \pi_0$, $X^{(t+1)} \mid X^{(0)}, \dots, X^{(t)} \sim Q(\cdot \mid X^{(t)})$ where $Q(y \mid x) = \mathbb{P}[X^{(t+1)} = y \mid X^{(t)} = x]$

**Stationary Distribution:** $\pi(y) = \int_x Q(y \mid x)\pi(x)d\mu(x)$ — MC satisfying DBE are reversible

↳ Detailed Balance: $\pi(x)Q(y \mid x) = \pi(y)Q(x \mid y)\ \forall x, y$ (sufficient for stationarity)

## Gibbs Sampler

Given param vector $\theta = (\theta_1, \dots, \theta_d)$

↳ Algorithm:
Initialize $\theta = \theta^{(0)}$
for $t = 1, \dots, T$:
 for $j = 1, \dots, d$:
  Sample $\theta_j \sim \lambda(\theta_j \mid \theta_{-j}, X)$ ↳ $\propto P(\theta_j \mid \theta_{P_a(j)}) \cdot \prod_{i: j \in P_a(i)} P(\theta_i \mid \theta_{P_a(i)})$
 Record $\theta^{(t)} = \theta$

Variations:
- Update one random coord $J^{(t)} \sim \text{Unif}\{1, \dots, d\}$
- Update coords in random order

Properties: • If $\theta^{(t)} \sim \lambda(\theta \mid X)$ then $\theta^{(t+1)} \sim \lambda(\theta \mid X)$
↳ Updating any coord in any order preserves posterior distro

## James-Stein Estimator

Some Definitions: $(Dh(x))_{ij} = \frac{\partial h_i}{\partial x_j}(x)$, $\|A\|_F = (\sum_{ij} A_{ij}^2)^{1/2}$

Estimator: $\delta_{JS,i}(X) = (1 - \frac{d-2}{\|X\|^2})X_i$ for $d \geq 3$ $\leftarrow$ Empirical Bayes Motivation: $\frac{d-2}{\|X\|^2}$ is UMVUE of $\zeta$

Paradox: $X_i \overset{iid}{\sim} N_d(\theta, \sigma^2 I_d)$, $i = 1, \dots, n$, $\theta \in \mathbb{R}^d$ (fixed), $\sigma^2 > 0$ known. Then, $\bar{X} = \frac{1}{n}\sum_{i=1}^n X_i$ is inadmissible as estimator of $\theta$ under squared error loss ($\delta_{JS}(X)$ better)
Non-Bayesian Gaussian Model ↗
↳ Useful for computing/estimating risk in Gaussian estimation problems

Lemma: Assume $X \sim N_d(\theta, \sigma^2 I_d)$, $h: \mathbb{R}^d \to \mathbb{R}^d$, $\mathbb{E}\|Dh(x)\|_F < \infty$: $\mathbb{E}[(X-\theta)^\top h(X)] = \sigma^2 \mathbb{E}[\text{tr}(Dh(X))] = \sigma^2 \sum_{i=1}^d \mathbb{E}\left[\frac{\partial h_i}{\partial x_i}(X)\right]$

SURE: $\hat{R}(X) = \sigma^2 d + \|h(X)\|^2 + 2\sigma^2 \text{tr}(Dh(X))$

## Minimax Risk

$\rightarrow$ Definition: $r^* = \inf_\delta \sup_\theta R(\theta; \delta)$

Minimax Estimator: $\delta^* \in \arg\inf_\delta \sup_\theta R(\theta; \delta)$ (i.e. achieves minimax risk)

Least Favorable Prior: $\Lambda^*$ where $r_{\Lambda^*} = \sup_\Lambda r_\Lambda$, $\sup_\theta R(\theta; \delta) \geq r^* \geq r_{\Lambda^*} \geq r_\Lambda$ (any $\delta$, any $\Lambda$)

Theorem: if $r_\Lambda = \sup_\theta R(\theta; \delta_\Lambda)$ w/ Bayes $\delta_\Lambda$, then: (1) $\delta_\Lambda$ minimax (2) Unique Bayes $\Rightarrow$ Unique minimax (3) $\Lambda$ least favorable

↳ avg risk = sup risk if $R(\theta; \delta_\Lambda)$ const and $\Lambda(\{\theta: R(\theta; \delta_\Lambda) = \max_\zeta R(\zeta; \delta_\Lambda)\}) = 1$

Least Favorable Sequence: sequence $\{\Lambda_n\}_{n \geq 1}$ is LF if $r_{\Lambda_n} \to \sup_\Lambda r_\Lambda$

Bounds: Upper: $r^* \leq \sup_\theta R(\theta; \delta)\ \forall \delta$
Lower: $r^* \geq \int R(\theta; \delta_\Lambda)\, d\Lambda(\theta)\ \forall \theta$

## Hypothesis Testing

Null: $H_0: \theta \in \Theta_0 \subseteq \Theta$ ↙ Simple hypothesis when single distribution
Alt: $H_1: \theta \in \Theta_1 \subseteq \Theta$

$R = \{x: \phi(x) = 1\}$ rejection region
$A = \{x: \phi(x) = 0\}$ acceptance region

Critical/Test Function: $\phi(x) = \begin{cases} 0 & \text{accept } H_0 \\ \pi \in (0,1) & \text{reject wp } \pi \\ 1 & \text{reject } H_0 \end{cases}$ $\rightarrow \phi$ is level-$\alpha$ test ($\alpha \in [0,1]$) if $\sup_{\theta \in \Theta_0} \beta_\phi(\theta) \leq \alpha$

Power Function: $\beta_\phi(\theta) = \mathbb{E}_\theta[\phi(X)] = \mathbb{P}_\theta[\text{Reject } H_0]$

$LR(x) = p_1(x)/p_0(x)$ where $p_1, p_0$ are densities of $\theta_1, \theta_0$

Likelihood Ratio Test (LRT): $\phi^*(x) = \begin{cases} 1 & LR(x) > c \\ \gamma & LR(x) = c \\ 0 & LR(x) < c \end{cases}$ where $c, \gamma$ chosen to make $\mathbb{E}_0[\phi^*(X)] = \alpha$

↳ Neyman Pearson Lemma: level-$\alpha$ LRT is optimal for testing $H_0: X \sim p_0$ vs. $H_1: X \sim p_1$ (simple hypotheses)

UMP: $\phi^*(x)$ is uniformly most powerful if for any other same-level test $\phi$ we have $\mathbb{E}_\theta[\phi^*] \geq \mathbb{E}_\theta[\phi]\ \forall \theta \in \Theta$ (Typically only exists for 1-sided testing in certain 1-param families)

Monotone Likelihood Ratios: $\mathcal{P} = \{P_\theta: \theta \in \Theta \subseteq \mathbb{R}\}$ has MLR if $\exists\ T(x)$ s.t. $\frac{p_{\theta_2}}{p_{\theta_1}}(x)$ is a non-decreasing function of $T(x)$ for any $\theta_1 < \theta_2$.

Theorem: Assume $\mathcal{P}$ has MLR, test $H_0: \theta \leq \theta_0$ vs. $H_1: \theta > \theta_0$ at level $\alpha \in (0,1)$. Let $\phi^*(x) = \begin{cases} 1 & T(x) > c \\ \gamma & T(x) = c \\ 0 & T(x) < c \end{cases}$, $c, \gamma$ chosen s.t. $\mathbb{E}_{\theta_0}[\phi^*(X)] = \alpha$
$\phi^*$ is LRT $\Rightarrow$ Then (1) $\phi^*$ is a UMP level-$\alpha$ test; and (2) If $\theta_1 < \theta_0$ then $\phi^*$ minimizes $\mathbb{E}_{\theta_1}\phi(X)$ among all $\phi$ s.t. $\mathbb{E}_{\theta_0}\phi(X) = \alpha$

One-Sided Test: $H_0: \theta \overset{\geq}{\leq} \theta_0$ vs $H_1: \theta \overset{<}{>} \theta_0$, often no UMP test exists

Stochastically Increasing: $T(X)$ is SI in $\theta$ if $\mathbb{P}_\theta[T(X) \leq t]$ is non-increasing in $\theta$, $\forall t$
↳ If $\phi(x)$ right-tailed on $T(X)$ and $T(X)$ SI, then $\mathbb{E}_\theta \phi(X)$ is increasing in $\theta$

Two-Sided Test: two sided hyps $H_0: \theta \in [\theta_1, \theta_2]$ vs $H_1: \theta \notin [\theta_1, \theta_2]$, then two-sided test rejects extreme values of $T(X)$:
$\phi(x) = \begin{cases} 1 & T(x) > c_2 \text{ or } T(x) < c_1 \\ 0 & T(x) \in (c_1, c_2) \\ \gamma_i & T(x) = c_i \end{cases}$

↳ Equal-Tailed Test: Let $\alpha_1 = \mathbb{P}_{\theta_0}[T < c_1] + \gamma_1 \mathbb{P}_{\theta_0}[T = c_1] = \alpha/2$
$\alpha_2 = \mathbb{P}_{\theta_0}[T > c_2] + \gamma_2 \mathbb{P}_{\theta_0}[T = c_2] = \alpha/2$

↳ Unbiased Test: $\phi(x)$ unbiased if $\inf_{\theta \in \Theta_1} \mathbb{E}_\theta \phi(X) \geq \alpha$ (ensure min $\beta_\phi(\theta) = \alpha$) $\rightarrow$ Choose $c_1, \gamma_1$ and $c_2, \gamma_2$ to solve: $\beta_\phi(\theta_0) = \alpha$, $\frac{d\beta_\phi}{d\theta}(\theta_0) = 0$

Theorem: Assume $X_i \overset{iid}{\sim} e^{\theta T(x) - A(\theta)}h(x)$ and two-tailed test. Then:
(1) Unbiased test on $\sum_i T(X_i)$ w/ level $\alpha$ is UMPU (UMP among all unbiased tests)
(2) If $\theta_1 < \theta_2$, UMPU test found by solving $c_i, \gamma_i$ s.t. $\mathbb{E}_{\theta_1}\phi = \mathbb{E}_{\theta_2}\phi = \alpha$
(3) If $\theta_1 = \theta_2 = \theta$, UMPU test found by solving $c_i, \gamma_i$ s.t. $\beta_\phi(\theta_0) = \alpha$ and $\frac{d\beta_\phi}{d\theta}(\theta_0) = \mathbb{E}_{\theta_0}[\sum_i T(X_i)(\phi(X) - \alpha)] = 0$

## p-value

Informal Definition: $p(x) = \mathbb{P}_{H_0}[T(X) \geq T(x)] = \sup_{\theta \in \Theta_0} \mathbb{P}_\theta[T(X) \geq T(x)]$

Definition: Assume we have $\phi_\alpha$ for each $\alpha$, $\sup_{\theta \in \Theta_0} \mathbb{E}_\theta \phi_\alpha(X) \leq \alpha$, and tests are monotone in $\alpha$. Then:
$p(x) = \inf\{\alpha: \phi_\alpha(x) = 1\}$ (= $\inf\{\alpha: x \in R_\alpha\}$ for non-randomized)

↳ For $\theta \in \Theta_0$, $\mathbb{P}_\theta[p(X) \leq \alpha] \leq \inf_{\tilde{\alpha} > \alpha} \underbrace{\mathbb{P}_\theta[\phi_{\tilde{\alpha}}(X) = 1]}_{\leq \tilde{\alpha}} \leq \alpha \Rightarrow$ p-value stochastically dominates Unif[0,1]

## Confidence Regions

Confidence Set: $C(X)$ is a $1-\alpha$ confidence set for $g(\theta)$ if coverage probability $= \mathbb{P}_\theta[C(X) \ni g(\theta)] \geq 1-\alpha,\ \forall \theta \in \Theta$ *$C(X)$ random, not $g(\theta)$

↳ $C(X)$ covers $g(\theta)$ if $C(X) \ni g(\theta)$; Confidence Level: $\inf_\theta \mathbb{P}_\theta[C(X) \ni g(\theta)]$

Duality of Tests and Confidence Sets:
(1) Given level-$\alpha$ test $\phi(x, a)$ of $H_0: g(\theta) = a$ vs $H_1: g(\theta) \neq a$, $\forall a \in g(\Theta)$. Then, we can construct CS for $g(\theta)$: $C(X) = \{a: \phi(X; a) < 1\}$
↳ $\mathbb{P}_\theta[C(X) \not\ni g(\theta)] = \mathbb{P}_\theta[\phi(X; g(\theta)) = 1] \leq \alpha\ \forall \theta$
(2) Given $1-\alpha$ CS $C(X)$ for $g(\theta)$. Then, can construct test $\phi(x)$ of $H_0: g(\theta) = a$ vs $H_1: g(\theta) \neq a$, $\phi(X) = 1\{a \notin C(X)\}$.
↳ $\mathbb{E}_\theta \phi(X) = \mathbb{P}_\theta[C(X) \not\ni g(\theta)] \leq \alpha\ \ \forall \theta$ s.t. $g(\theta) = a$

↳ Procedure in (1) is called "Inverting the test"

Confidence Interval: A CS $C(X)$ is a CI if it's in the form $C(X) = [C_1(X), C_2(X)]$
↳ Lower Confidence Bound $LCB(X) = [C_1(X), \infty)$, Upper Confidence Bound $UCB(X) = (-\infty, C_2(X)]$ — Obtained by inverting a one-sided test; If test UMP, called Uniformly Most Accurate (UMA)
↳ Obtain CI by inverting two-sided test; if UMPU then called UMAU

## Nuisance Parameters

Setup: $\mathcal{P} = \{P_{\theta,\lambda}: (\theta, \lambda) \in \Omega\}$, $H_0: \theta \in \Theta_0$ vs $H_1: \theta \in \Theta_1$
Parameter of Interest ↗ ↖ Nuisance Parameter: extra unknown param that might affect stuff

Multiparameter Exp. Families
- Assume $X \sim p_{\theta,\lambda}(x) = \exp\{\theta^\top T(x) + \lambda^\top U(x) - A(\theta,\lambda)\} \cdot h(x)$ where $\theta \in \mathbb{R}^s$, $\lambda \in \mathbb{R}^r$ both unknown
- To test $H_0: \theta \in \Theta_0$ vs $H_1: \theta \in \Theta_1$, we perform the following: *Idea: condition on $U(X)$ to elim. $\lambda$

1. Sufficiency Reduction: $(T(X), U(X)) \sim q_{\theta,\lambda}(t,u) = \exp\{\theta^\top t + \lambda^\top u - A(\theta,\lambda)\} \cdot g(t,u)$
2. Condition on $U(X)$: $q_\theta(t \mid u) = q_{\theta,\lambda}(t,u) / \int q_{\theta,\lambda}(z,u)\,dz = \exp\{\theta^\top t - B_u(\theta)\} \cdot g(t,u)$ where $B_u(\theta) = \int e^{\theta^\top z} g(z,u)\,dz$
3. Conditional Test: Test $H_0: \theta \in \Theta_0$ vs $H_1: \theta \in \Theta_1$ in model $\mathcal{Q}_u = \{q_\theta(t \mid u): \theta \in \Theta\}$

Note: if $s = 1$, $\mathcal{Q}_u$ has MLR in $T$

Theorem:
1) For $H_0: \theta \leq \theta_0$ vs $H_1: \theta > \theta_0$, use UMPU $\phi^*(X) = \psi(T(X); U(X))$ where $\psi(t; u) = \begin{cases} 1 & t > c(u) \\ \gamma(u) & t = c(u) \\ 0 & t < c(u) \end{cases}$
w/ $c(u), \gamma(u)$ chosen st. $\mathbb{E}_{\theta_0}[\phi^*(X) \mid U(X) = u] = \alpha$
2) For $H_0: \theta = \theta_0$ vs $H_1: \theta \neq \theta_0$, use UMPU $\phi^*(X) = \psi(T(X); U(X))$ where $\psi(t; u) = \begin{cases} 1 & t \notin [c_1(u), c_2(u)] \\ \gamma_i(u) & t = c_i(u) \\ 0 & t \in (c_1(u), c_2(u)) \end{cases}$
w/ $c_i(u), \gamma_i(u)$ chosen st. $\mathbb{E}_{\theta_0}[\phi^*(X) \mid U(X) = u] = \alpha$, $\mathbb{E}_{\theta_0}[T(X)(\phi^*(X) - \alpha) \mid U(X) = u] = 0$

## Change of Basis

## Asymptotics

$X_n \xrightarrow{P} c$ if $\mathbb{P}[\|X_n - c\| > \varepsilon] \to 0,\ \forall \varepsilon > 0$
$X_n \xrightarrow{d} X$ if $\mathbb{E} f(X_n) \to \mathbb{E} f(X)\ \forall$ bdd, cts $f: \mathcal{X} \to \mathbb{R}$

$Df(x) = \begin{pmatrix} -\nabla f_1(x)- \\ \vdots \\ -\nabla f_k(x)- \end{pmatrix}$

Consistency: $\delta_n(X_n)$ consistent for $g(\theta)$ if $\delta_n(X_n) \xrightarrow{P_\theta} g(\theta)$, i.e. $\mathbb{P}_\theta[\|\delta_n(X_n) - g(\theta)\| > \varepsilon] \to 0$

LLN: if $\mathbb{E}|X_i| < \infty$, $\mathbb{E}X_i = \mu$, then $\bar{X}_n \xrightarrow{P} \mu$ ($\bar{X}_n \xrightarrow{a.s.} \mu$)

CLT: if $\mathbb{E}X = \mu \in \mathbb{R}^d$, $\text{Var}(X_n) = \Sigma < \infty$, then $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \Sigma)$

Delta Method: if $\sqrt{n}(X_n - \mu) \Rightarrow N_d(0, \Sigma)$ and $f: \mathbb{R}^d \to \mathbb{R}^k$ diff at $x = \mu$, then $\sqrt{n}(f(X_n) - f(\mu)) \Rightarrow N_k(0, Df(\mu)\Sigma Df(\mu)^\top)$ $= N(0, Df(\mu)^2 \Sigma Df(\mu))$ if $k = 1$
↳ In general, can do higher-order Taylor expansion: if $f'(\mu) = 0$, we have $n(f(X_n) - f(\mu)) \Rightarrow f''(\mu)/2\,(\sqrt{n}(X_n - \mu))^2 \approx f''(\mu)\sigma^2/2 \cdot \chi_1^2$

## Maximum Likelihood Estimation

$\rightarrow$ Definition: for dominated $\mathcal{P} = \{P_\theta: \theta \in \Theta\}$, MLE is $\hat{\theta}_{MLE}(X) = \arg\max_{\theta \in \Theta} p_\theta(X) = \arg\max_{\theta \in \Theta} \ell(\theta; X)$

Asymptotic Efficiency: Assume $X_1, \dots, X_n \overset{iid}{\sim} p_\theta(x)$, where $p_\theta$ smooth in $\theta$. Let $\ell_1(\theta; X_i) = \log p_\theta(X_i)$, $\ell_n(\theta; X) = \sum_{i=1}^n \ell_1(\theta; X_i)$
$J_1(\theta) = \text{Var}_\theta(\nabla \ell_1(\theta; X_i)) = -\mathbb{E}_\theta[\nabla^2 \ell_1(\theta; X_i)]$
$J_n(\theta) = \text{Var}_\theta(\nabla \ell_n(\theta; X_i)) = nJ_1(\theta)$
Then, estimator $\hat{\theta}_n$ is asymptotically efficient if $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{P_\theta} N(0, J_1(\theta)^{-1})$

↳ Delta Method for $g(\theta)$: $\sqrt{n}(g(\hat{\theta}_n) - g(\theta)) \xrightarrow{P_\theta} N(0, \nabla g(\theta)^\top J_1(\theta)^{-1} \nabla g(\theta))$
↳ $\hat{\theta}_{MLE}$ is asymptotically gaussian and efficient

KL-Divergence: $D_{KL}(\theta_0 \| \theta) = \mathbb{E}_{\theta_0}[\log p_{\theta_0}(X)/p_\theta(X)]$

Asymptotic Distro of MLE: (*) Assume $X_1, \dots, X_n \overset{iid}{\sim} p_{\theta_0}$ for $\theta_0 \in \Theta \subseteq \mathbb{R}^d$, $\hat{\theta}_n \in \arg\max_{\theta \in \Theta} \ell_n(\theta; X)$, $\hat{\theta}_n \xrightarrow{P} \theta_0$, etc., and $\mathbb{E}_{\theta_0} \nabla \ell_1(\theta_0; X) = 0$, $\text{Var}_{\theta_0} \nabla \ell_1(\theta_0; X) = -\mathbb{E}_{\theta_0} \nabla^2 \ell_1(\theta_0; X) \succ 0$. Then we have:
$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N_d(0, J_1(\theta_0)^{-1})$

## Likelihood-Based Inference

Same assumptions as (*) directly above. We then have
$\frac{1}{\sqrt{n}} \nabla \ell_n(\theta_0) \Rightarrow N_d(0, J_1(\theta_0))$, $-\frac{1}{n}\nabla^2 \ell_n(\theta_0) \xrightarrow{P} J_1(\theta_0)$, $\sqrt{n}(\hat{\theta}_n - \theta_0) \Rightarrow N_d(0, J_1(\theta_0)^{-1})$

Wald-Type Confidence: Assume estimator $\hat{J}_n \succ 0$ s.t. $\frac{1}{n}\hat{J}_n \xrightarrow{P} J_1(\theta_0) \succ 0$. Then, to test $H_0: \theta = \theta_0$ vs $H_1: \theta \neq \theta_0$ we reject if $\|\hat{J}_n^{1/2}(\hat{\theta}_n - \theta_0)\|_2^2 > \chi_d^2(\alpha)$. This gives confidence ellipsoid.
↳ Option 1: $\hat{J}_n = nJ_1(\hat{\theta}_n) = n\text{Var}_{\hat{\theta}_n}(\nabla \ell_n(\hat{\theta}_n; X))$
↳ Option 2: $\hat{J}_n = -\nabla^2 \ell_n(\hat{\theta}_n; X)$ $\leftarrow$ Preferred!

Univariate Wald Interval: $\theta_j \in C_j = \hat{\theta}_{n,j} \pm \sqrt{(\hat{J}_n^{-1})_{jj}} \cdot z_{\alpha/2}$

Score Test: For $H_0: \theta = \theta_0$ vs $H_1: \theta \neq \theta_0$, use score as test stat; then $J_n(\theta_0)^{-1/2} \nabla \ell_n(\theta_0; X) \overset{H_0}{\Rightarrow} N_d(0, I_d)$.
We reject if $\|J_n(\theta_0)^{-1/2} \nabla \ell_n(\theta_0; X)\|^2 > \chi_d^2(\alpha)$ *Asymptotically same as Wald Test
↳ No quadratic approx, no MLE, no need to estimate Fisher info at $\theta_0$

GLRT w/ simple null: for $H_0: \theta = \theta_0$ vs $H_1: \theta \neq \theta_0$, use test stat $2(\ell_n(\hat{\theta}_n; X) - \ell_n(\theta_0; X)) \overset{P_0}{\Rightarrow} \chi_d^2$

GLRT w/ composite null or nuisance params: use thm to get $2(\ell_n(\hat{\theta}_n) - \ell_n(\hat{\theta}_0)) \Rightarrow \chi_{d-d_0}^2$, $\hat{\theta}_0 = \arg\max_{\theta \in \Theta_0} \ell_n(\theta; X)$

Asymptotic Relative Efficiency: $\hat{\theta}_n^{(i)}$ w/ $i = 1, 2$ estimators w/ $d = 1$ and $\sqrt{n}(\hat{\theta}_n^{(i)} - \theta_0) \xrightarrow{d} N(0, \sigma_i^2)$. Then ARE of $\hat{\theta}^{(2)}$ wrt $\hat{\theta}^{(1)}$ is $\sigma_1^2 / \sigma_2^2$

## Miscellaneous

- $\text{Bias}_\theta(\delta(X)) = \mathbb{E}_\theta[\delta(X)] - \theta$, $MSE(\theta; \delta) = \mathbb{E}_\theta\|\delta(X) - \theta\|^2 = \|\text{Bias}_\theta(\delta(X))\|_2^2 + \text{tr}(\text{Var}_\theta(\delta(X)))$